

# **TASK 1 – MULTIMETER**

## **1. PC Power Supply Problem**

A multimeter can be used to check the voltage coming from the power supply. This helps to know if the power supply is giving the correct voltage to the computer.

## **2. Laptop Charger Problem**

If a laptop charger is not working, a multimeter can be used to check its output voltage. This helps to find out if the charger is working properly or not.

## **3. Smartphone Charging Problem**

If a smartphone is not charging properly, a multimeter can be used to check the voltage from the charger or charging circuit. This can help to find the possible power problem.

# **TASK 2 – POST CARD**

A POST card is used to check the hardware when a computer starts. It displays POST codes that help identify possible hardware problems. The codes can give an idea about issues with components such as RAM, CPU or motherboard. The POST card helps the user identify the possible hardware problem.

# TASK 3 – SOFTWARE DIAGNOSTIC TOOL

## Using HWMonitor

HWMonitor is a software tool used to check the condition of computer hardware. It can show information like temperature, voltage and other system details.

### Steps:

1. Download and install HWMonitor.
2. Open HWMonitor on the computer.
3. Check the main dashboard.
4. Look at the available temperature, voltage and system information.
5. Take a screenshot of the dashboard.
6. Crop or blur any personal information before adding the screenshot to the assignment.

| CPUID HWMonitor           |                |                |                |
|---------------------------|----------------|----------------|----------------|
| File View Tools Help      |                |                |                |
| Sensor                    | Value          | Min            | Max            |
| MSI                       |                |                |                |
| USB Drive 2               |                |                |                |
| Utilization               |                |                |                |
| Space (f:)                | 39 %           | 39 %           | 39 %           |
| NVIDIA GeForce GTX 1080   |                |                |                |
| Temperatures              |                |                |                |
| GPU                       | 53 °C (127 °F) | 53 °C (127 °F) | 54 °C (129 °F) |
| Utilization               |                |                |                |
| GPU                       | 0 %            | 0 %            | 49 %           |
| Memory                    | 45 %           | 44 %           | 45 %           |
| Frame Buffer              | 9 %            | 9 %            | 35 %           |
| Video Engine              | 0 %            | 0 %            | 0 %            |
| Bus Interface             | 0 %            | 0 %            | 0 %            |
| Performance               |                |                |                |
| Power Limit               | 0              | 0              | 0              |
| Temperature Limit         | 0              | 0              | 0              |
| Reliable Voltage Limit    | 0              | 0              | 0              |
| Operational Voltage Limit | 0              | 0              | 0              |
| Utilization Limit         | 1              | 1              | 1              |
| SLI Synchronisation Limit | 0              | 0              | 0              |
| BIF0_9                    |                |                |                |
| Voltages                  |                |                |                |
| Current Voltage           | 16.598 V       | 16.597 V       | 16.598 V       |
| Capacities                |                |                |                |
| Designed Capacity         | 79344 mWh      | 79344 mWh      | 79344 mWh      |
| Full Charge Capacity      | 77904 mWh      | 77904 mWh      | 77904 mWh      |
| Current Capacity          | 76651 mWh      | 76651 mWh      | 76651 mWh      |
| Levels                    |                |                |                |
| Wear Level                | 2 %            | 2 %            | 2 %            |
| Charge Level              | 98 %           | 98 %           | 98 %           |
| Intel Core i7             |                |                |                |

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## TASK 4 – COMPARISON OF TROUBLESHOOTING TOOLS

| Tool                 | What it Checks                                           | Example Problem Detected        |
|----------------------|----------------------------------------------------------|---------------------------------|
| Multimeter           | Checks voltage, current and resistance                   | Power supply or charger problem |
| POST Card            | Checks hardware during computer startup using POST codes | RAM, CPU or motherboard problem |
| Software Diagnostics | Checks hardware condition, temperature and system status | Overheating or storage problem  |

## CONCLUSION

In this assignment, I learned about some useful tools that can help in computer troubleshooting.

I learned how a multimeter can be used to check voltage and other electrical values when there is a power-related problem. I also learned about a POST card, which can show codes during computer startup and help find hardware problems.

I also learned how software diagnostic tools like HWMonitor can show useful information about the computer, such as temperature and voltage.

Finally, I compared the multimeter, POST card and software diagnostic tools and understood that each tool is useful for finding different types of problems.

Overall, this assignment helped me understand some basic troubleshooting tools and how they can be used to find hardware and system problems.